

ATM

ATM stands for Automated Teller Machine. It is an electronic banking outlet that allows customers to perform various financial transactions such as cash withdrawals, deposits, funds transfer, and checking their account balance. ATMs are available 24 hours a day, 7 days a week, and are usually located in public areas such as shopping malls, convenience stores, and banks.

To use an ATM, a customer typically inserts their bank card into the machine and enters their Personal Identification Number (PIN). They can then choose from a range of options, such as withdrawing cash or transferring money between accounts. The ATM will then dispense cash or complete the requested transaction.

ATMs have become an integral part of modern banking, providing customers with convenient and easy access to their funds. They have also evolved over time, with many newer machines offering additional features such as printing mini statements, paying bills, and even accepting deposits of cash and checks.

OBJECTIVE: -

- To provide customers with a secure and easy way to access their bank accounts and perform transactions.
- To allow customers to withdraw cash, check their account balance, and transfer money.
- To provide customers with a convenient way to deposit money into their account.

WORKING OF ATM

ATMs (Automated Teller Machines) allow customers to perform various financial transactions without the need to visit a bank. Here are the steps involved in the working of an ATM:

Card Authentication: The first step is to insert your bank card into the ATM. The machine reads the magnetic strip or chip on your card and verifies that the card is valid and has not been reported lost or stolen.

PIN Entry: After the card is authenticated, you will be prompted to enter your Personal Identification Number (PIN) to access your account. The PIN is a secret code that only the cardholder knows, and it is used to ensure that only authorized individuals can access the account.

Transaction Selection: Once your account is authenticated, the ATM displays a menu of available transactions. You can select the desired transaction, such as withdrawing cash, depositing money, or checking your account balance.

Transaction Processing: After you have selected the transaction, the ATM processes the request and communicates with your bank's computer system to complete the transaction. This process usually takes only a few seconds.

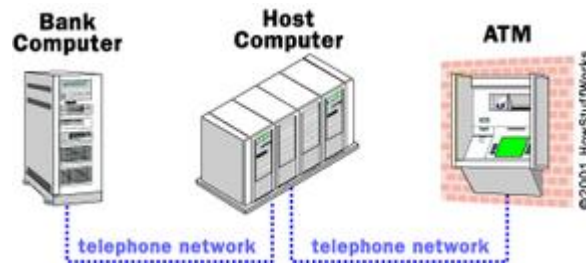
Dispensing Cash: If you have requested a cash withdrawal, the ATM will dispense the cash in the amount you specified. The cash is usually dispensed in a mix of denominations.

Transaction Receipt: The ATM then prints a transaction receipt for you to confirm that the transaction was completed. The receipt shows the details of the transaction, such as the amount withdrawn, the account balance, and any fees charged.

End of Transaction: Once the transaction is complete, the ATM returns your bank card, and the screen displays a message indicating that the transaction is finished. You can then choose to perform another transaction or end the session.

ATMs have become an essential part of modern banking, providing customers with quick and convenient access to their funds. They have evolved over time,

with many newer machines offering additional features such as depositing cash and checks, paying bills, and transferring funds between accounts



Here's a basic flow chart of an embedded system in an ATM:

- User inserts the ATM card.
- The card reader checks the card for validity and authenticity.
- If the card is valid, the user is prompted to enter their Personal Identification Number (PIN).
- The user enters their PIN using the keypad.
- The ATM verifies the PIN and grants access to the account.
- The user selects the transaction they want to perform, such as cash withdrawal, account balance check, or funds transfer.
- The ATM communicates with the bank's computer system to process the transaction.
- If the transaction involves dispensing cash, the ATM dispenses the requested amount of cash.
- The ATM prints a receipt for the transaction, which includes details such as the amount withdrawn, account balance, and any fees charged.
- The user retrieves their card and any cash dispensed.
- The ATM returns to the initial state and waits for the next user.

This is a simplified flow chart of the ATM embedded system, and the actual process may involve additional steps or different sequences of operations depending on the specific ATM and bank's configuration. However, the basic operations of card reading, PIN verification, transaction processing, and cash dispensing remain the same.

COMPONENTS IN EMBEDDED SYSTEM OF ATM

The embedded system in an ATM (Automated Teller Machine) is composed of several hardware and software components that work together to enable the machine's functionality. Some of the key components of the embedded system in an ATM are:

- **CPU (Central Processing Unit):** The CPU is the brain of the ATM and performs all the processing tasks. It executes the software program that controls the ATM and manages all the hardware components.
- **Card Reader:** The card reader reads the magnetic stripe or chip on the ATM card and sends the card data to the CPU for processing. It also checks the card for validity and authenticity.
- **Keypad:** The keypad allows the user to input their PIN and select the desired transaction. It contains several keys, including numeric keys and function keys for navigating the ATM menus.
- **Display:** The display provides visual feedback to the user and displays information such as the transaction status, account balance, and instructions for using the ATM.
- **Cash Dispenser:** The cash dispenser is the mechanism that dispenses the cash to the user during a withdrawal transaction. It includes a storage cassette for holding the cash and a dispenser mechanism for dispensing the correct amount of cash.
- **Receipt Printer:** The receipt printer prints a receipt for the user after a transaction is completed. The receipt contains details such as the transaction amount, account balance, and any fees charged.
- **Network Interface:** The network interface connects the ATM to the bank's computer system and allows the ATM to communicate with the bank for transaction processing and other functions.
- **Operating System:** The ATM operating system is the software that manages the hardware components and provides the user interface for interacting with the ATM.

All these components work together to provide a reliable, secure, and user-friendly ATM experience. The specific components and their configurations may vary depending on the ATM model and the bank's requirements.

System Design:

- **Hardware:** The hardware of the embedded system for an ATM includes a touch screen, a card reader, a cash dispenser, a receipt printer, and a secure keypad. The hardware components are connected to a central processing unit (CPU), which controls the overall functionality of the system.
- **Software:** The software of the embedded system for an ATM includes an operating system, a user interface, and a transaction management system. The user interface is designed to be user-friendly and intuitive, allowing customers to easily perform transactions. The transaction management system is responsible for verifying the customer's identity, managing the flow of funds, and updating the customer's account information.
- **Security:** The embedded system for an ATM is designed with security in mind. The system includes features such as encryption of sensitive data, two-factor authentication, and a secure keypad for entering passwords. In addition, the system is regularly updated with the latest security patches to protect against potential security threats.

Implementation:

- **Installation:** The embedded system for an ATM is installed at bank branches, retail outlets, and other convenient locations.
- **Training:** Bank employees are trained on how to use the system and how to assist customers with transactions.
- **Maintenance:** The embedded system for an ATM is regularly maintained to ensure that it is functioning properly and to fix any issues that may arise.

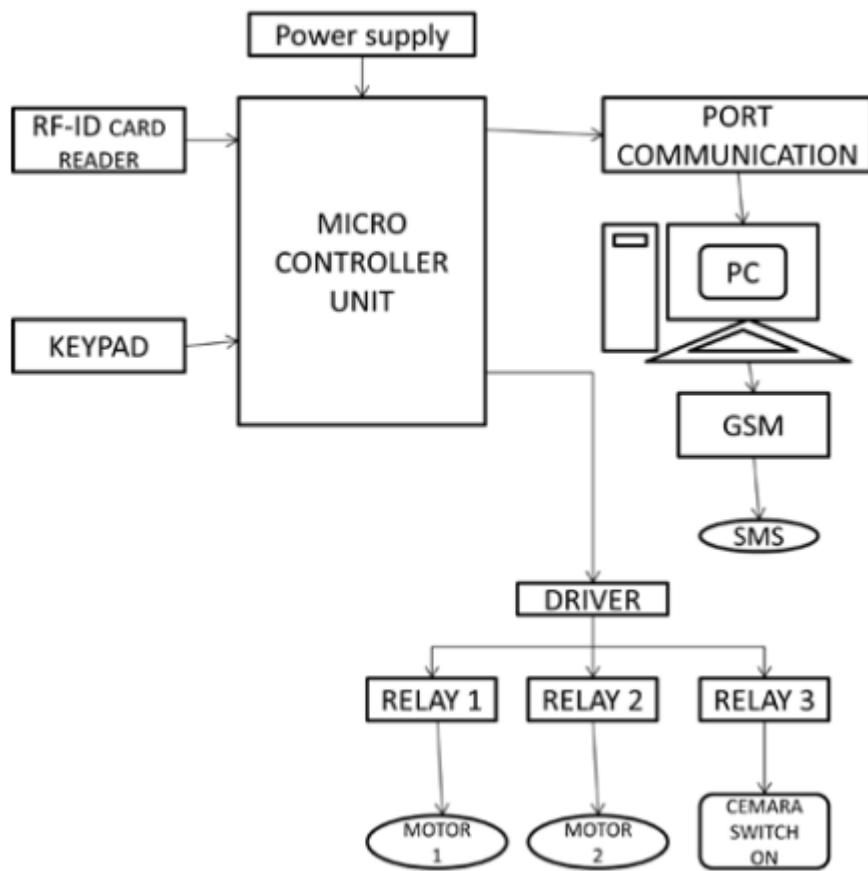
PIN DIAGRAM USED IN EMBEDDED SYSTEM OF ATM

The pin diagram used in an embedded system of an ATM (Automated Teller Machine) may vary depending on the specific hardware and microcontroller used. However, here is a general overview of the pins typically used in an ATM system:

- **Power supply pins:** These pins provide power to the microcontroller and other components of the ATM system.
- **Input/output (I/O) pins:** These pins are used to interface the microcontroller with other devices such as the keypad, display, card reader, and cash dispenser. The keypad and card reader typically use digital I/O pins, while the display and cash dispenser may use serial or parallel communication protocols.
- **Communication pins:** These pins are used to communicate with external devices such as the bank's server, the card processing network, and the ATM's network interface. These pins may include serial communication interfaces such as UART, SPI, or I2C.
- **Interrupt pins:** These pins are used to trigger an interrupt in the microcontroller when a specific event occurs, such as a button press or card insertion.
- **Clock and timing pins:** These pins are used to synchronize the operation of the microcontroller and other components in the ATM system.
- **Reset pin:** This pin is used to reset the microcontroller and other components in the ATM system.

Again, the specific pin diagram used in an ATM system may vary depending on the hardware and microcontroller used. It's important to refer to the datasheet and documentation of the specific components in use for an accurate pin diagram.

BLOCK DIAGRAM: -



HARDWARE REQUIREMENTS: -

- GSM Modem
- Personal computer
- Microcontroller
- RFID Reader
 - Driver
- Relay
- Motor
- Keypad
- Power supply

SOFTWARE REQUIREMENTS: -

Real-time operating system (RTOS): The embedded system of an ATM typically uses a real-time operating system (RTOS) to manage hardware and software components. An RTOS is designed for real-time applications and is optimized for tasks such as process scheduling, interrupt handling, and memory management.

Device drivers: The embedded system of an ATM requires device drivers to interface with various hardware components such as the card reader, keypad, and cash dispenser. These drivers allow the RTOS to communicate with the hardware and control its operation.

Middleware: The ATM's embedded system may require middleware components to handle communication between hardware and software components. Middleware provides a layer of abstraction between the RTOS and the application software, making it easier to develop and maintain the software.

Application software: The embedded system of an ATM requires application software to handle various types of transactions, such as cash withdrawals, deposits, and balance inquiries. This software typically interfaces with the user through a touch screen or keypad and communicates with the bank's servers to process transactions.

Security software: An ATM's embedded system requires several security software components to protect it from physical and electronic attacks. These components may include software to detect and prevent card skimming, software to monitor the ATM for tampering or vandalism, and software to detect and prevent unauthorized access to the ATM's internal components.

Communication software: The embedded system of an ATM requires software to communicate with the bank's servers over the internet or a dedicated ATM network. This software is responsible for encrypting and decrypting data, handling network protocols, and ensuring secure communication.